Results and Analysis (EP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

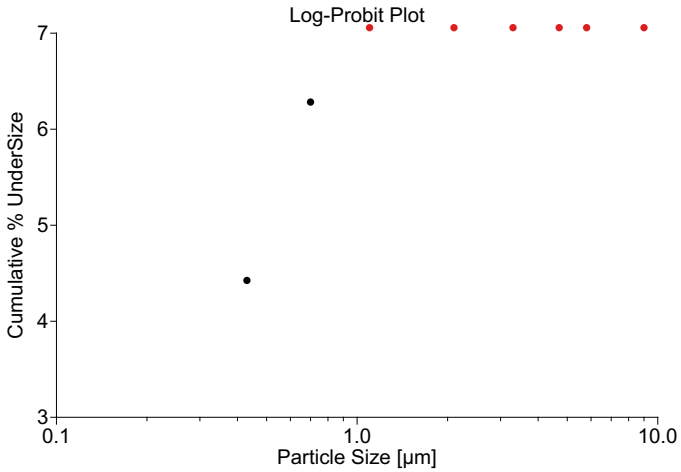
Other Info : Validation V3.1 - [File CI 1]

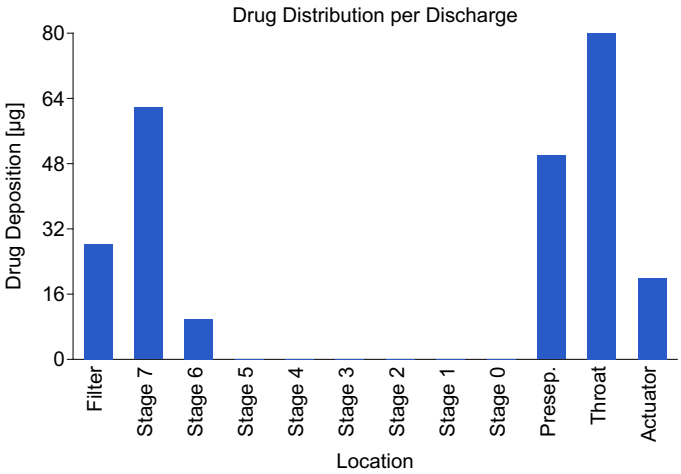
Run Number =	1	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	1.12300+/-2.12300

Cut-off Diameter [µm]	Mass of active ingredient deposited per discharge [µg]	Cumulative mass of active ingredient deposited per discharge [µg]	Cumulative fraction of active ingredient deposited per discharge [%]
0.43000	28.26928	28.26928	28.26928
0.70000	61.74669	90.01597	90.01597
1.10000	9.85133	99.86730	99.86730
2.10000	0.13270	100.00000	100.00000
3.30000	0.00000	100.00000	100.00000
4.70000	0.00000	100.00000	100.00000
5.80000	0.00000	100.00000	100.00000
9.00000	0.00000	100.00000	100.00000
---	0.00000	100.00000	100.00000

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 230.00000
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	100.00000
MMAD [µm] =	0.50000
GSD =	1.30000





Log-Probit Graph Regression:

Intercept =	NA
Slope =	NA
R^2 =	NA
n =	1

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

Results and Analysis (EP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

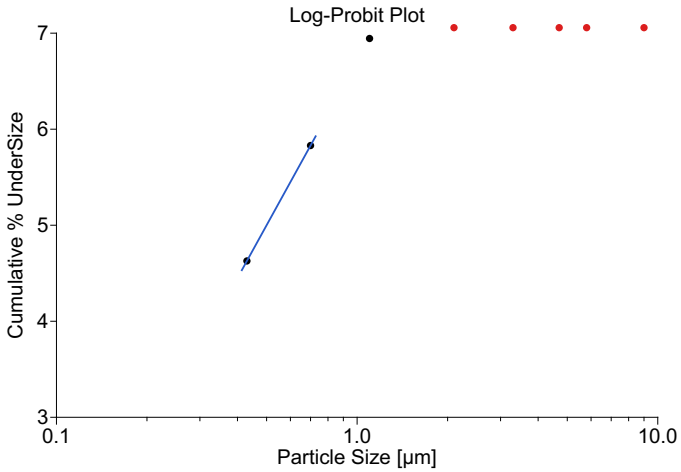
Other Info : Validation V3.1 - [File CI 1]

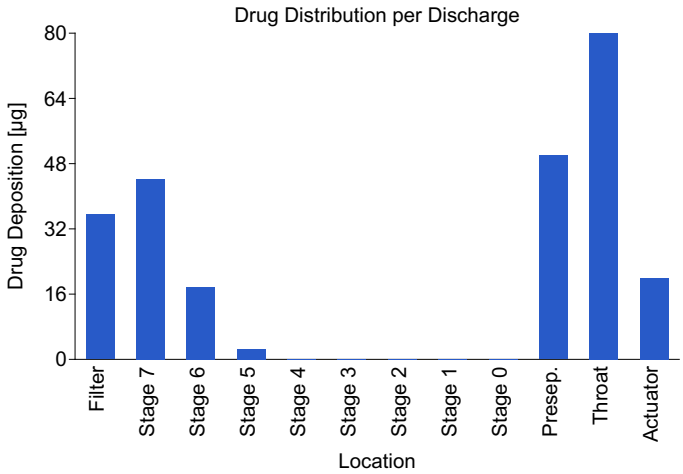
Run Number =	2	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	3.12300+/-4.12300

Cut-off Diameter [µm]	Mass of active ingredient deposited per discharge [µg]	Cumulative mass of active ingredient deposited per discharge [µg]	Cumulative fraction of active ingredient deposited per discharge [%]
0.43000	35.49557	35.49557	35.49557
0.70000	44.17304	79.66861	79.66862
1.10000	17.74008	97.40869	97.40870
2.10000	2.57125	99.97994	99.97995
3.30000	0.01989	99.99983	99.99984
4.70000	0.00016	99.99999	100.00000
5.80000	0.00000	99.99999	100.00000
9.00000	0.00000	99.99999	100.00000
---	0.00000	99.99999	100.00000

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 229.99999
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	99.99999
MMAD [µm] =	0.50000
GSD =	1.50000





Log-Probit Graph Regression:

Intercept =	6.70951
Slope =	5.67887
R^2 =	1.00000
n =	2

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

Results and Analysis (EP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

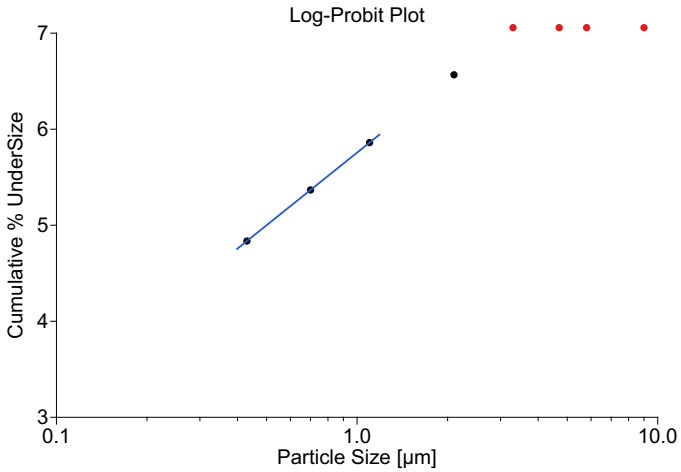
Other Info : Validation V3.1 - [File CI 1]

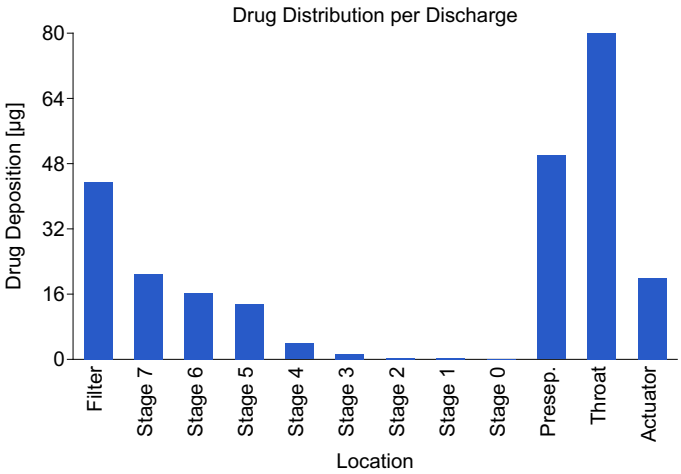
Run Number =	3	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	5.12300+/-6.12300

Cut-off Diameter [µm]	Mass of active ingredient deposited per discharge [µg]	Cumulative mass of active ingredient deposited per discharge [µg]	Cumulative fraction of active ingredient deposited per discharge [%]
0.43000	43.46288	43.46288	43.46288
0.70000	20.86405	64.32693	64.32692
1.10000	16.19707	80.52400	80.52399
2.10000	13.61078	94.13478	94.13477
3.30000	3.89274	98.02752	98.02751
4.70000	1.24905	99.27657	99.27656
5.80000	0.34970	99.62627	99.62626
9.00000	0.29333	99.91960	99.91959
---	0.08041	100.00001	100.00000

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 230.00001
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	99.40136
MMAD [µm] =	0.50000
GSD =	2.50000





Log-Probit Graph Regression:

Intercept =	5.75647
Slope =	2.51294
R^2 =	1.00000
n =	3

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

Results and Analysis (EP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

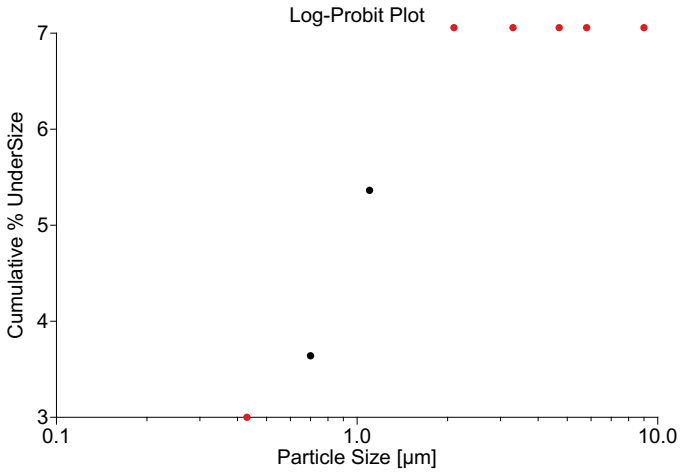
Other Info : Validation V3.1 - [File CI 1]

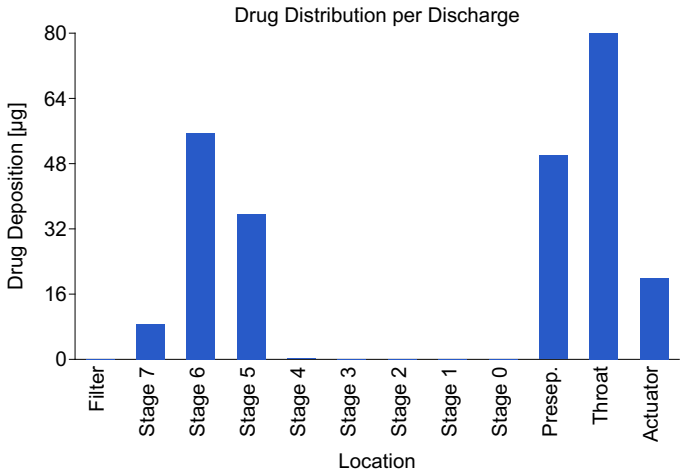
Run Number =	4	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	7.12300+/-8.12300

Cut-off Diameter [µm]	Mass of active ingredient deposited per discharge [µg]	Cumulative mass of active ingredient deposited per discharge [µg]	Cumulative fraction of active ingredient deposited per discharge [%]
0.43000	0.06482	0.06482	0.06482
0.70000	8.63515	8.69997	8.69997
1.10000	55.48003	64.18000	64.18000
2.10000	35.58572	99.76572	99.76572
3.30000	0.23401	99.99973	99.99973
4.70000	0.00027	100.00000	100.00000
5.80000	0.00000	100.00000	100.00000
9.00000	0.00000	100.00000	100.00000
---	0.00000	100.00000	100.00000

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 230.00000
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	100.00000
MMAD [µm] =	1.00000
GSD =	1.30000





Log-Probit Graph Regression:

Intercept =	NA
Slope =	NA
R^2 =	NA
n =	1

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

Results and Analysis (EP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

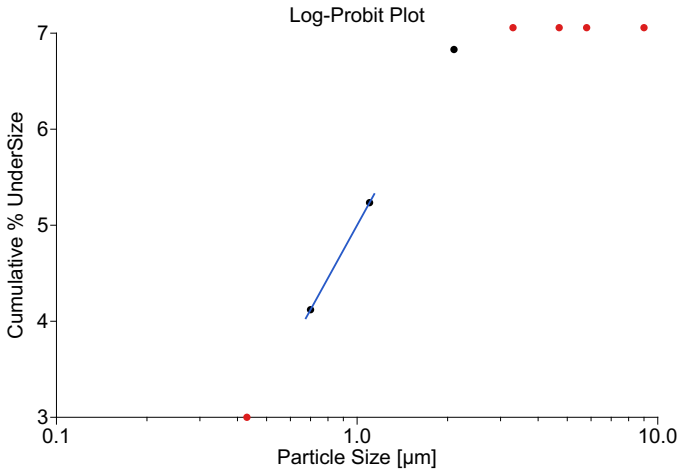
Other Info : Validation V3.1 - [File CI 1]

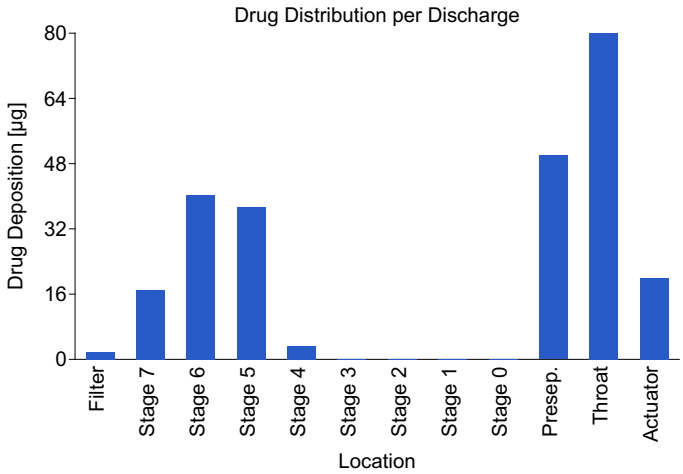
Run Number =	5	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	9.12300+/-10.12300

Cut-off Diameter [µm]	Mass of active ingredient deposited per discharge [µg]	Cumulative mass of active ingredient deposited per discharge [µg]	Cumulative fraction of active ingredient deposited per discharge [%]
0.43000	1.86947	1.86947	1.86947
0.70000	17.08247	18.95194	18.95194
1.10000	40.34010	59.29204	59.29205
2.10000	37.34429	96.63633	96.63634
3.30000	3.20197	99.83830	99.83831
4.70000	0.15494	99.99324	99.99325
5.80000	0.00603	99.99927	99.99928
9.00000	0.00072	99.99999	100.00000
---	0.00000	99.99999	100.00000

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 229.99999
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	99.99927
MMAD [µm] =	1.00000
GSD =	1.50000





Log-Probit Graph Regression:

Intercept =	5.00000
Slope =	5.67887
R^2 =	1.00000
n =	2

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

Results and Analysis (EP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

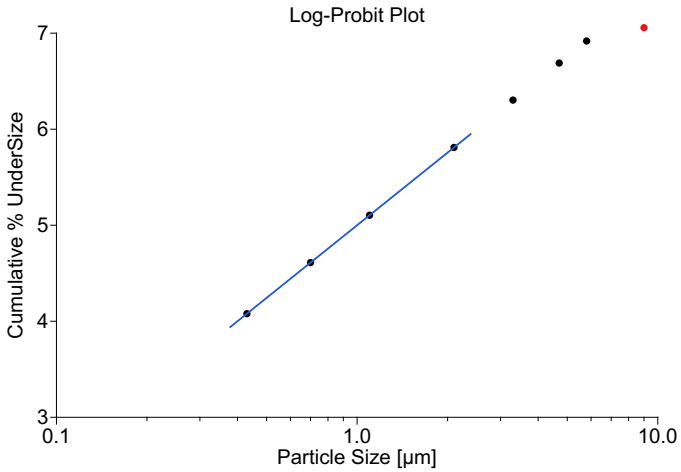
Other Info : Validation V3.1 - [File CI 1]

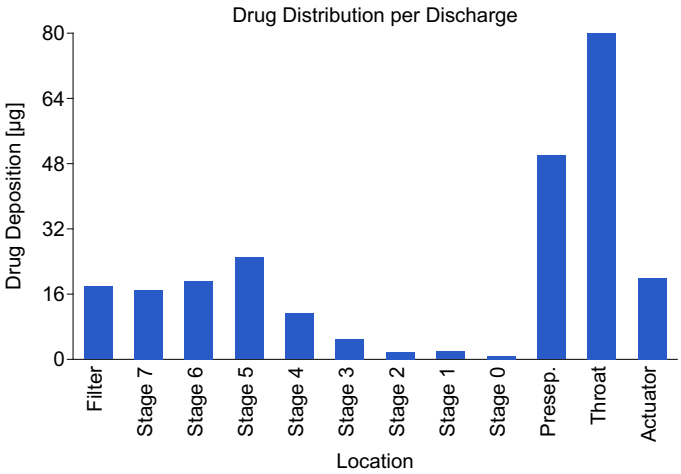
Run Number =	6	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	11.12300+/-12.12300

Cut-off Diameter [µm]	Mass of active ingredient deposited per discharge [µg]	Cumulative mass of active ingredient deposited per discharge [µg]	Cumulative fraction of active ingredient deposited per discharge [%]
0.43000	17.85063	17.85063	17.85063
0.70000	17.00357	34.85420	34.85420
1.10000	19.28802	54.14222	54.14222
2.10000	24.95267	79.09489	79.09489
3.30000	11.27629	90.37118	90.37118
4.70000	5.06730	95.43848	95.43848
5.80000	1.80882	97.24730	97.24730
9.00000	1.92836	99.17566	99.17566
---	0.82434	100.00000	100.00000

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 230.00000
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	96.04960
MMAD [µm] =	1.00000
GSD =	2.50000





Log-Probit Graph Regression:

Intercept =	5.00000
Slope =	2.51294
R^2 =	1.00000
n =	4

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

Results and Analysis (EP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

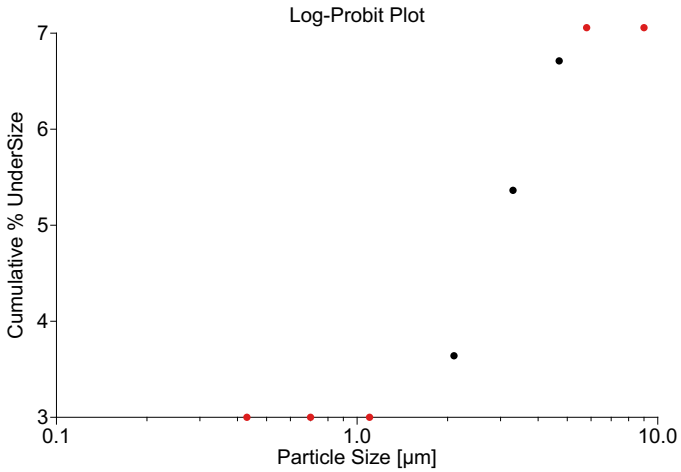
Other Info : Validation V3.1 - [File CI 1]

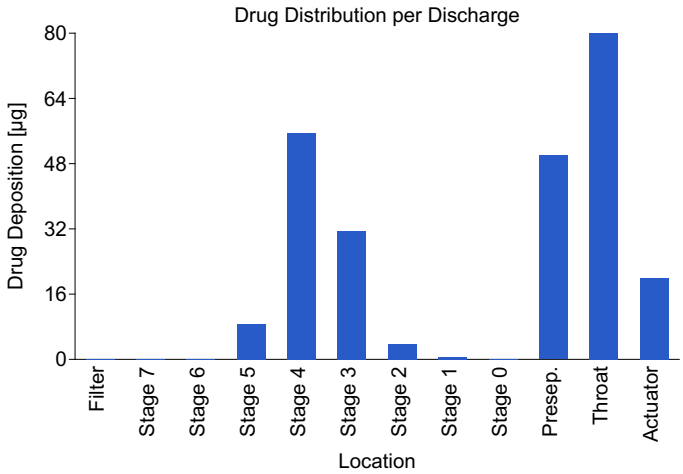
Run Number =	7	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	13.12300+/-14.12300

Cut-off Diameter [µm]	Mass of active ingredient deposited per discharge [µg]	Cumulative mass of active ingredient deposited per discharge [µg]	Cumulative fraction of active ingredient deposited per discharge [%]
0.43000	0.00000	0.00000	0.00000
0.70000	0.00000	0.00000	0.00000
1.10000	0.00656	0.00656	0.00656
2.10000	8.69341	8.69997	8.69997
3.30000	55.48003	64.18000	64.17999
4.70000	31.46753	95.64753	95.64752
5.80000	3.75344	99.40097	99.40096
9.00000	0.59763	99.99860	99.99859
---	0.00141	100.00001	100.00000

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 230.00001
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	97.42332
MMAD [µm] =	3.00000
GSD =	1.30000





Log-Probit Graph Regression:

Intercept =	NA
Slope =	NA
R^2 =	NA
n =	1

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

Results and Analysis (EP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

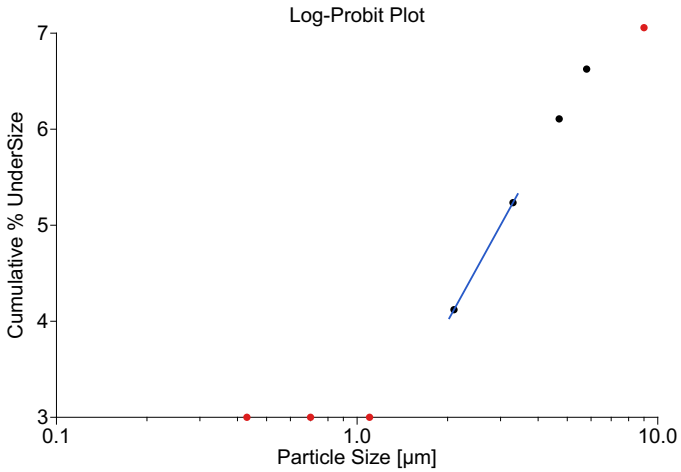
Other Info : Validation V3.1 - [File CI 1]

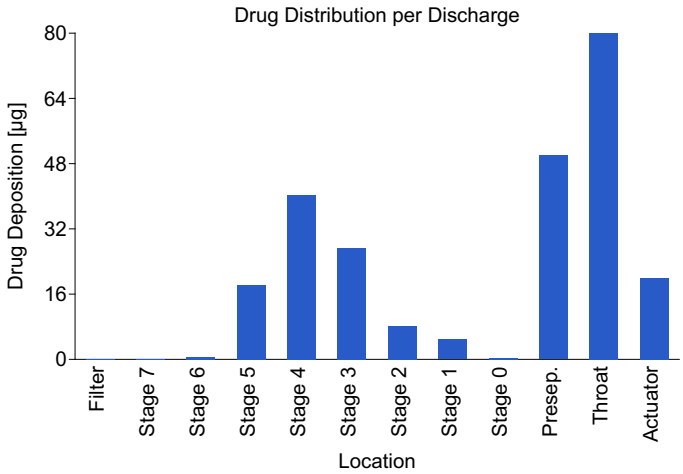
Run Number =	8	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	15.12300+/-16.12300

Cut-off Diameter [µm]	Mass of active ingredient deposited per discharge [µg]	Cumulative mass of active ingredient deposited per discharge [µg]	Cumulative fraction of active ingredient deposited per discharge [%]
0.43000	0.00008	0.00008	0.00008
0.70000	0.01650	0.01658	0.01658
1.10000	0.65063	0.66721	0.66721
2.10000	18.28473	18.95194	18.95194
3.30000	40.34010	59.29204	59.29204
4.70000	27.29861	86.59065	86.59065
5.80000	8.21080	94.80145	94.80145
9.00000	4.86164	99.66309	99.66309
---	0.33691	100.00000	100.00000

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 230.00000
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	89.61384
MMAD [µm] =	3.00000
GSD =	1.50000





Log-Probit Graph Regression:

Intercept =	2.29049
Slope =	5.67887
R^2 =	1.00000
n =	2

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

Results and Analysis (EP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

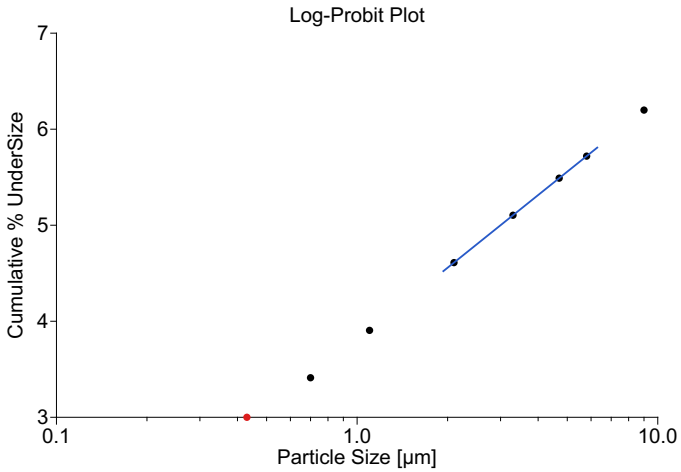
Other Info : Validation V3.1 - [File CI 1]

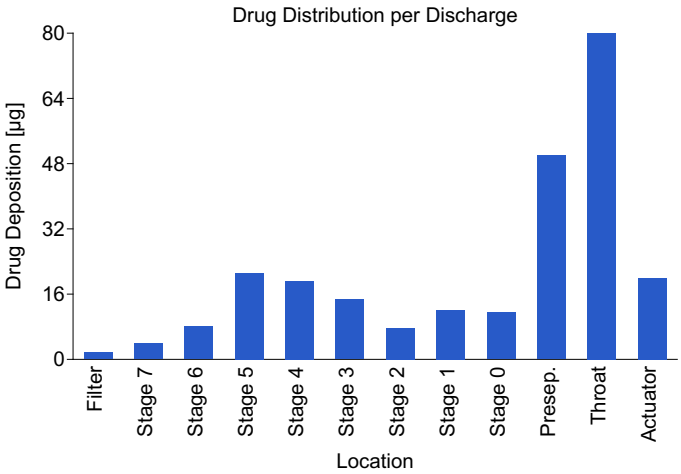
Run Number =	9	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	17.12300+/-18.12300

Cut-off Diameter [µm]	Mass of active ingredient deposited per discharge [µg]	Cumulative mass of active ingredient deposited per discharge [µg]	Cumulative fraction of active ingredient deposited per discharge [%]
0.43000	1.70009	1.70009	1.70009
0.70000	3.91154	5.61163	5.61163
1.10000	8.06507	13.67670	13.67670
2.10000	21.17751	34.85421	34.85420
3.30000	19.28802	54.14223	54.14222
4.70000	14.64984	68.79207	68.79206
5.80000	7.61544	76.40751	76.40749
9.00000	12.06568	88.47319	88.47317
---	11.52683	100.00002	100.00000

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 230.00002
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	71.14048
MMAD [µm] =	3.00000
GSD =	2.50000





Log-Probit Graph Regression:

Intercept =	3.80102
Slope =	2.51294
R^2 =	1.00000
n =	4

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

Results and Analysis (EP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

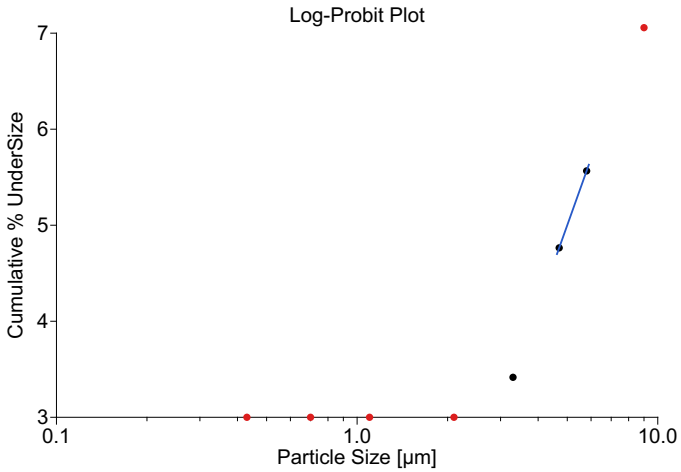
Other Info : Validation V3.1 - [File CI 1]

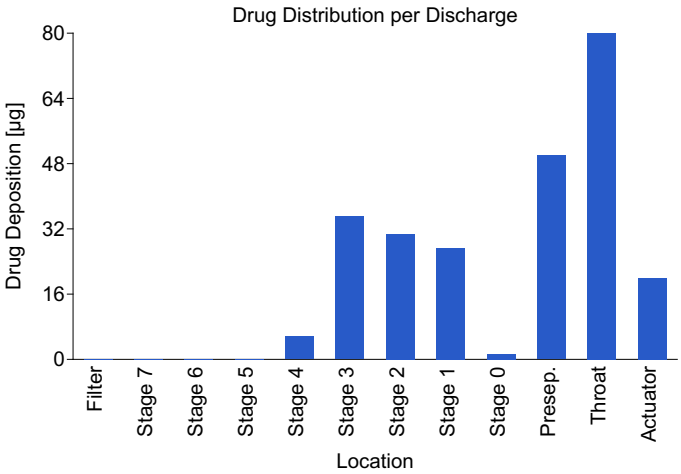
Run Number =	10	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	19.12300+/-20.12300

Cut-off Diameter [µm]	Mass of active ingredient deposited per discharge [µg]	Cumulative mass of active ingredient deposited per discharge [µg]	Cumulative fraction of active ingredient deposited per discharge [%]
0.43000	0.00000	0.00000	0.00000
0.70000	0.00000	0.00000	0.00000
1.10000	0.00000	0.00000	0.00000
2.10000	0.04724	0.04724	0.04724
3.30000	5.61546	5.66270	5.66270
4.70000	35.01522	40.67792	40.67792
5.80000	30.74226	71.42018	71.42019
9.00000	27.32639	98.74657	98.74658
---	1.25342	99.99999	100.00000

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 229.99999
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	49.99999
MMAD [µm] =	5.00000
GSD =	1.30000





Log-Probit Graph Regression:

Intercept =	-1.13437
Slope =	8.77629
R^2 =	1.00000
n =	2

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

Results and Analysis (EP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

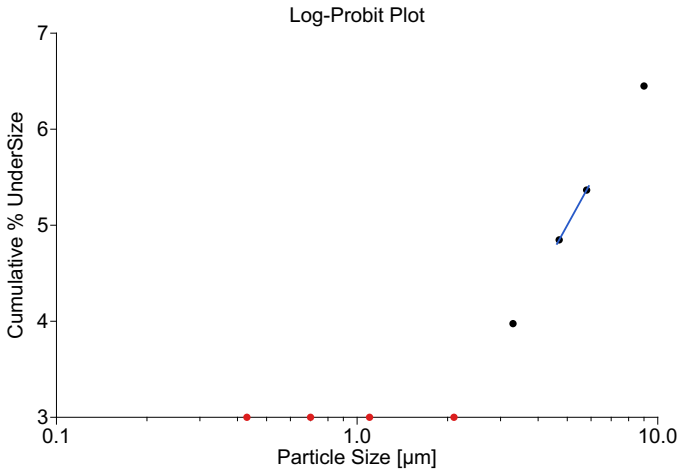
Other Info : Validation V3.1 - [File CI 1]

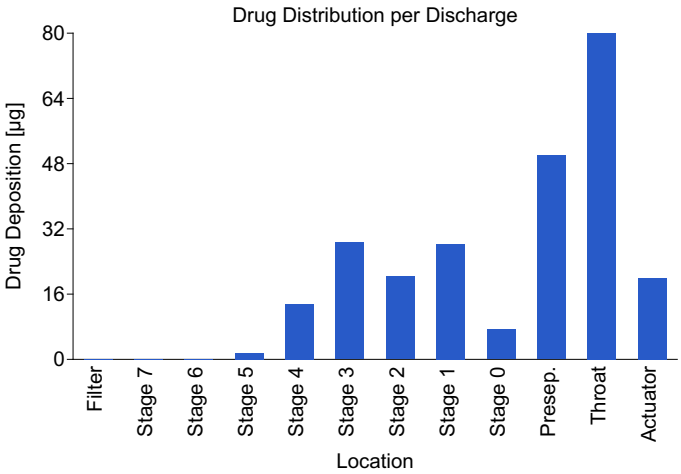
Run Number =	11	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	21.12300+/-22.12300

Cut-off Diameter [µm]	Mass of active ingredient deposited per discharge [µg]	Cumulative mass of active ingredient deposited per discharge [µg]	Cumulative fraction of active ingredient deposited per discharge [%]
0.43000	0.00000	0.00000	0.00000
0.70000	0.00006	0.00006	0.00006
1.10000	0.00935	0.00941	0.00941
2.10000	1.61027	1.61968	1.61968
3.30000	13.65350	15.27318	15.27318
4.70000	28.66237	43.93555	43.93555
5.80000	20.34802	64.28357	64.28357
9.00000	28.35877	92.64234	92.64234
---	7.35766	100.00000	100.00000

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 230.00000
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	50.00000
MMAD [µm] =	5.00000
GSD =	1.50000





Log-Probit Graph Regression:

Intercept =	1.03064
Slope =	5.67887
R^2 =	1.00000
n =	2

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

Results and Analysis (EP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

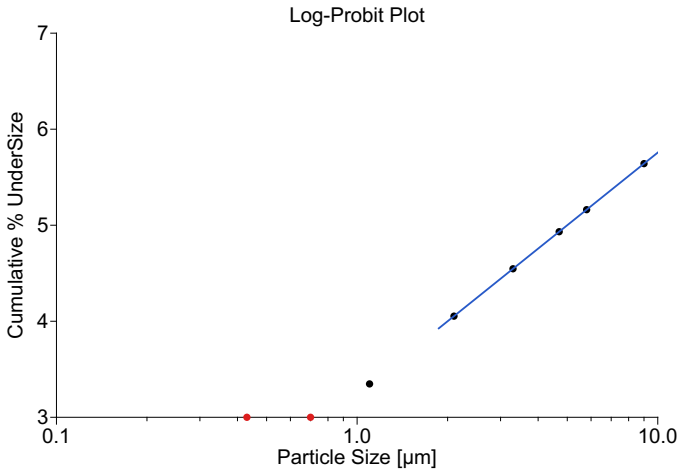
Other Info : Validation V3.1 - [File CI 1]

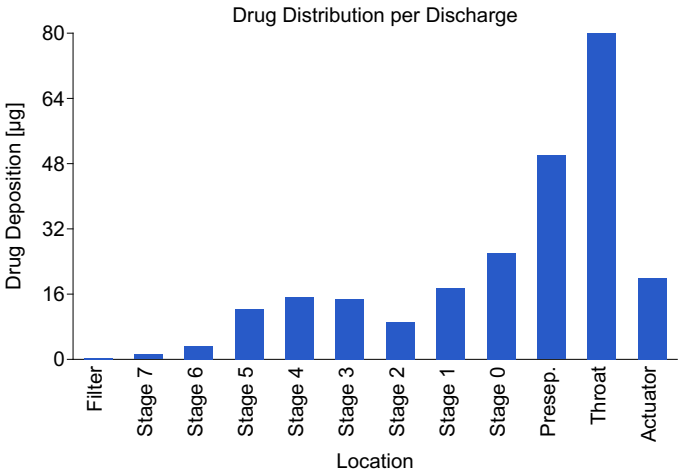
Run Number =	12	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	12.12300+/-13.12300

Cut-off Diameter [µm]	Mass of active ingredient deposited per discharge [µg]	Cumulative mass of active ingredient deposited per discharge [µg]	Cumulative fraction of active ingredient deposited per discharge [%]
0.43000	0.37082	0.37082	0.37082
0.70000	1.22390	1.59472	1.59472
1.10000	3.32738	4.92210	4.92210
2.10000	12.26614	17.18824	17.18824
3.30000	15.32207	32.51031	32.51031
4.70000	14.79775	47.30806	47.30806
5.80000	9.12582	56.43388	56.43389
9.00000	17.50573	73.93961	73.93962
---	26.06038	99.99999	100.00000

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 229.99999
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	50.00000
MMAD [µm] =	5.00000
GSD =	2.50000





Log-Probit Graph Regression:

Intercept =	3.24353
Slope =	2.51294
R^2 =	1.00000
n =	5

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

Results and Analysis (EP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

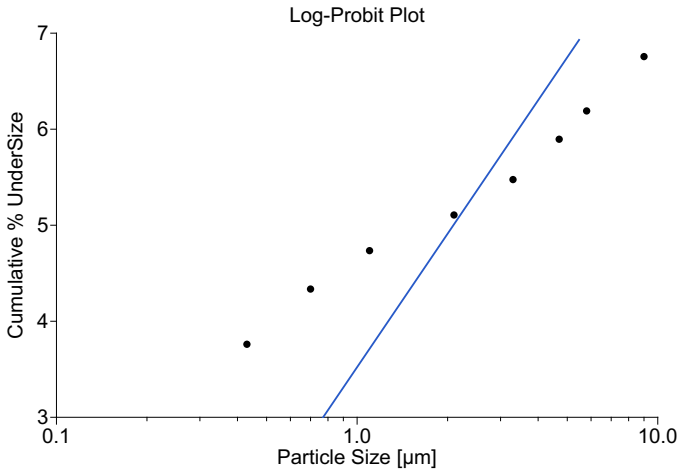
Other Info : Validation V3.1 - [File CI 1]

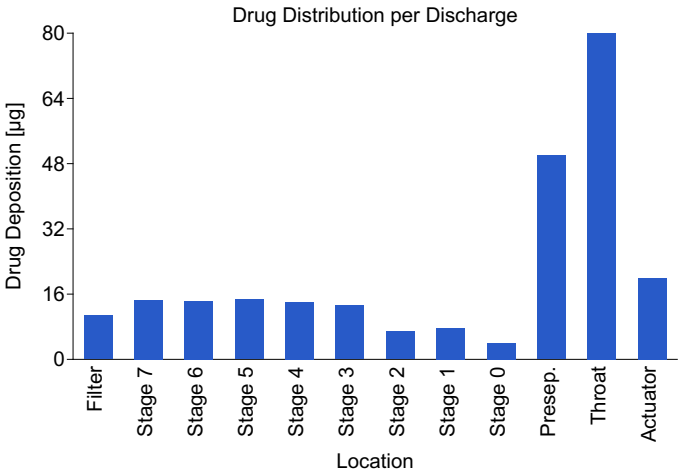
Run Number =	Mean	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	11.20633+/-12.20633

Cut-off Diameter [µm]	Mass of active ingredient deposited per discharge [µg]	Cumulative mass of active ingredient deposited per discharge [µg]	Cumulative fraction of active ingredient deposited per discharge [%]
0.43000	10.75697	10.75697	10.75697
0.70000	14.55475	25.31172	25.31172
1.10000	14.24630	39.55802	39.55802
2.10000	14.68973	54.24775	54.24774
3.30000	14.02701	68.27475	68.27475
4.70000	13.19692	81.47167	81.47167
5.80000	6.83003	88.30170	88.30170
9.00000	7.74485	96.04655	96.04655
---	3.95345	100.00000	100.00000

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 230.00000
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	83.63565
MMAD [µm] =	2.37500
GSD =	1.76667





Log-Probit Graph Regression:

Intercept =	3.52192
Slope =	4.61595
R^2 =	1.00000
n =	2.41667

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

Results and Analysis (USP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

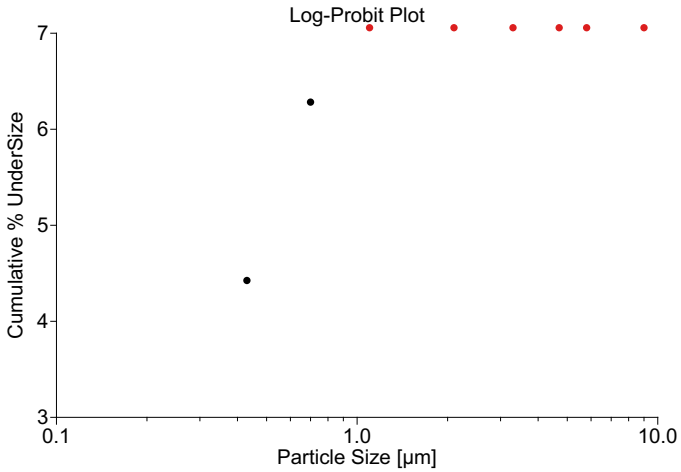
Other Info : Validation V3.1 - [File CI 1]

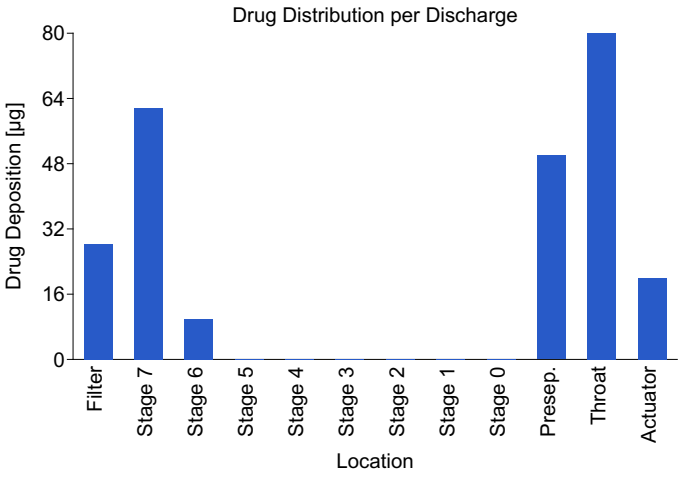
Run Number =	1	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	1.12300+/-2.12300

Stage	Mass A [µg]	Mass B [µg]	Cum. % of mass less than stated diameter	<D50 [µm]
Filter	28.26928	28.26928	28.26928	0.43000
Stage 7	61.74669	61.74669	90.01597	0.70000
Stage 6	9.85133	9.85133	99.86730	1.10000
Stage 5	0.13270	0.13270	100.00000	2.10000
Stage 4	0.00000	0.00000	100.00000	3.30000
Stage 3	0.00000	0.00000	100.00000	4.70000
Stage 2	0.00000	0.00000	100.00000	5.80000
Stage 1	0.00000	0.00000	100.00000	9.00000
Stage 0	0.00000	0.00000	100.00000	---
Presep.	50.00000			
M.A./I.P.	80.00000			
Sum of Masses	230.00000 (A)	100.00000 (B)		

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 230.00000
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	100.00000
Fine Particle Fraction, Less than or Equal to 5.00000 µm [%] =	43.47826
MMAD [µm] =	0.50000
GSD =	1.30000





Log-Probit Graph Regression:

Intercept =	NA
Slope =	NA
R^2 =	NA
n =	1

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

Results and Analysis (USP Method) for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

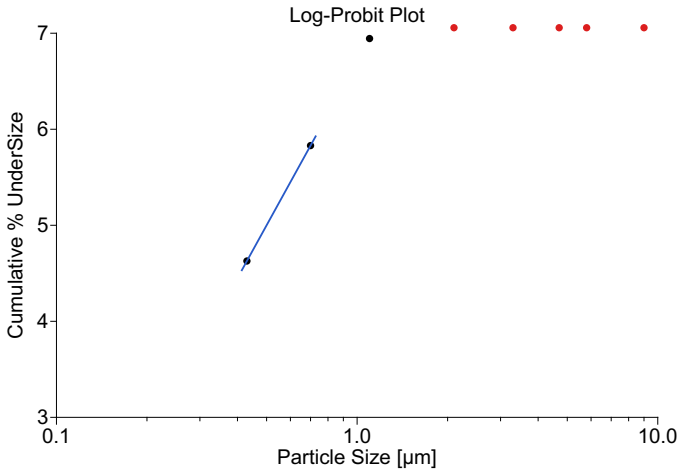
Other Info : Validation V3.1 - [File CI 1]

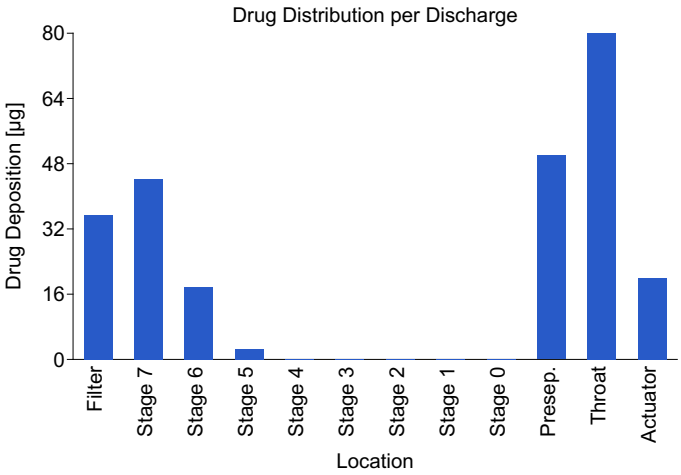
Run Number =	2	Sampling Flow rate (l/min) =	28.30000
Number of discharges =	1	Shot Weight, mean +/- sd [mg] =	3.12300+/-4.12300

Stage	Mass A [µg]	Mass B [µg]	Cum. % of mass less than stated diameter	<D50 [µm]
Filter	35.49557	35.49557	35.49557	0.43000
Stage 7	44.17304	44.17304	79.66862	0.70000
Stage 6	17.74008	17.74008	97.40870	1.10000
Stage 5	2.57125	2.57125	99.97995	2.10000
Stage 4	0.01989	0.01989	99.99984	3.30000
Stage 3	0.00016	0.00016	100.00000	4.70000
Stage 2	0.00000	0.00000	100.00000	5.80000
Stage 1	0.00000	0.00000	100.00000	9.00000
Stage 0	0.00000	0.00000	100.00000	---
Presep.	50.00000			
M.A./I.P.	80.00000			
Sum of Masses	229.99999 (A)	99.99999 (B)		

(1) Total mass of active ingredient, less device deposition per discharge [µg] = 229.99999
(2) Mean delivered dose determined during testing for uniformity of delivered dose [µg] = 230.00000
(3) Total mass of active substance within 75-125% of mean delivered dose : 100.0% - Result within specification.
(4) Total mass of active substance within 80-120% of mean delivered dose : 100.0% - Result within specification.
(5) Total mass of active substance within 85-115% of mean delivered dose : 100.0% - Result within specification.

Less than or Equal to 5.00000 µm [µg] =	99.99999
Fine Particle Fraction, Less than or Equal to 5.00000 µm [%] =	43.47826
MMAD [µm] =	0.50000
GSD =	1.50000





Log-Probit Graph Regression:

Intercept =	6.70951
Slope =	5.67887
R^2 =	1.00000
n =	2

File Name : FILE CI 1.CPL

Data Input by: Date:

Data Checked by: Date:

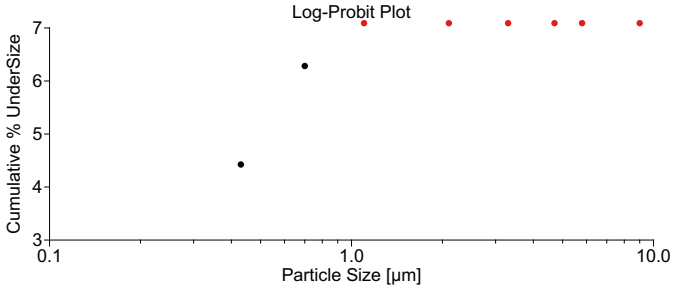
Graphical Summary for Cascade Impactor

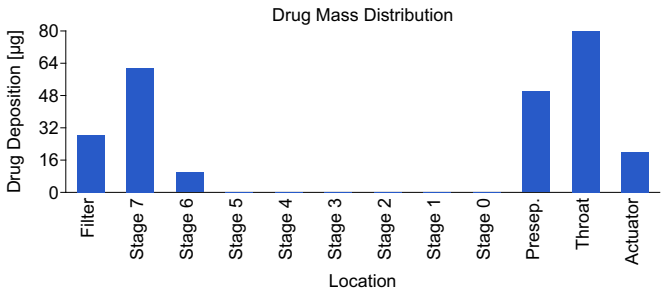
Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

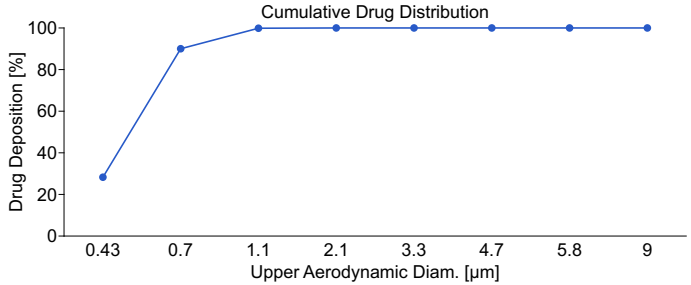
Other Info : Validation V3.1 - [File CI 1]

File Name: FILE CI 1.CPL

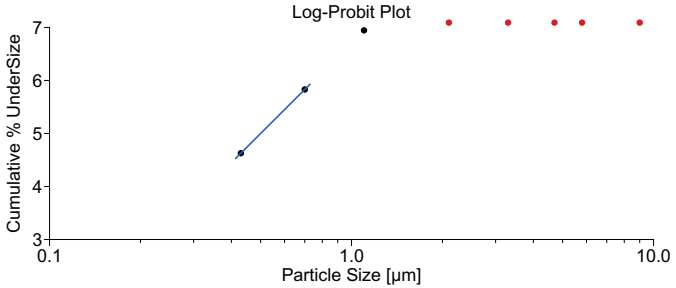
Run 1 Sampling Flow Rate (l/min) = 28.30

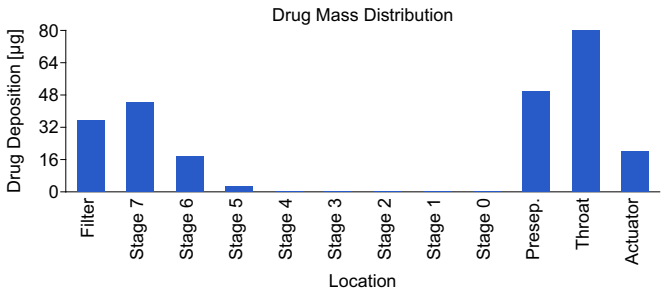


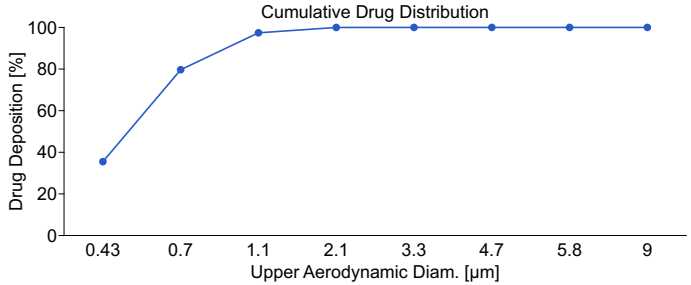




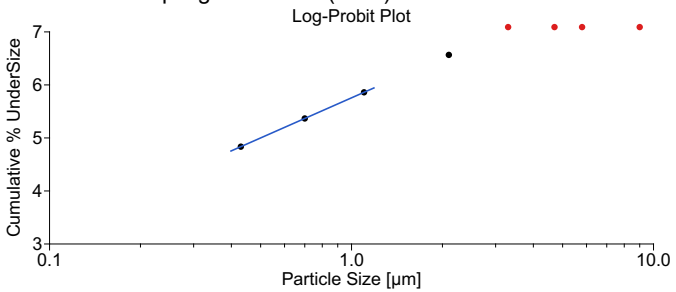
Run 2 Sampling Flow Rate (l/min) = 28.30

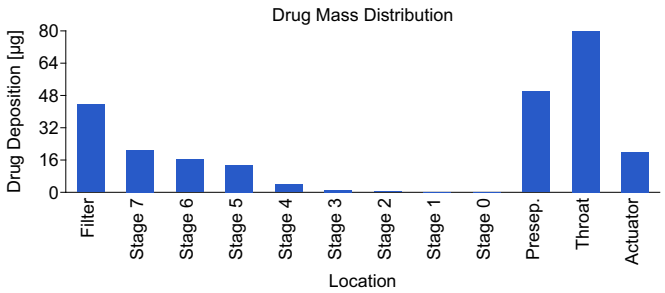


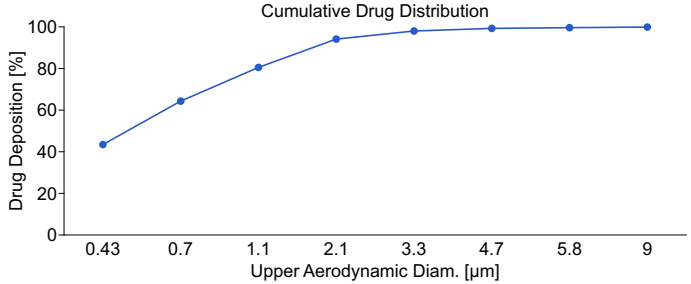




Run 3 Sampling Flow Rate (l/min) = 28.30







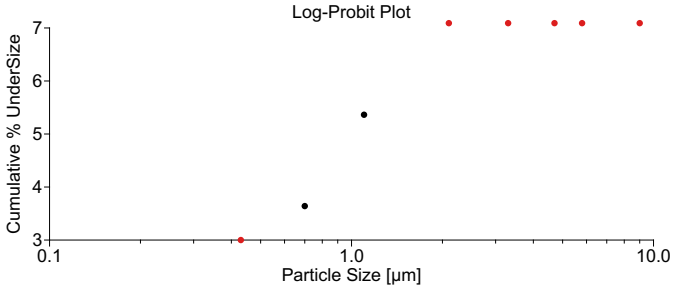
Graphical Summary for Cascade Impactor

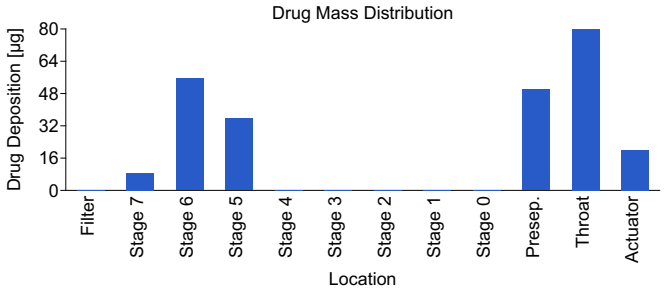
Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

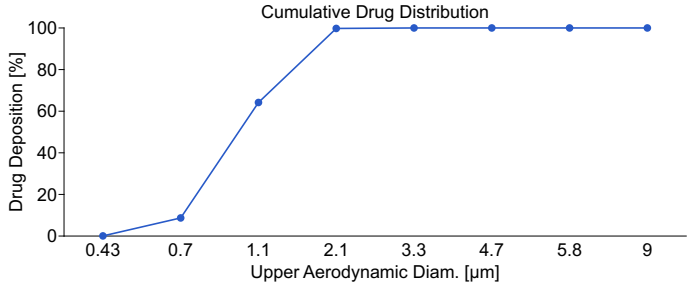
Other Info : Validation V3.1 - [File CI 1]

File Name: FILE CI 1.CPL

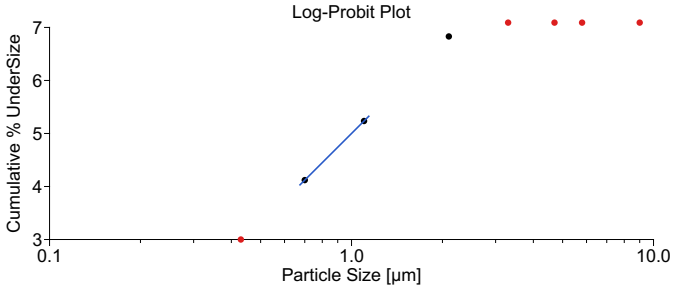
Run 4 Sampling Flow Rate (l/min) = 28.30

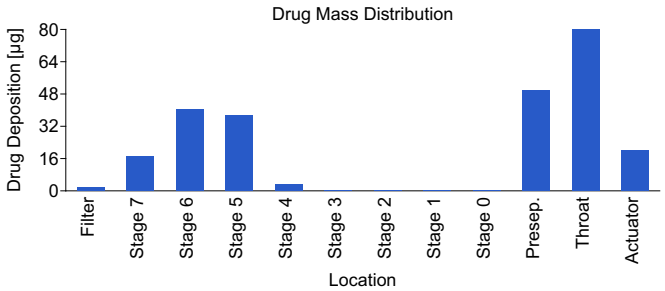


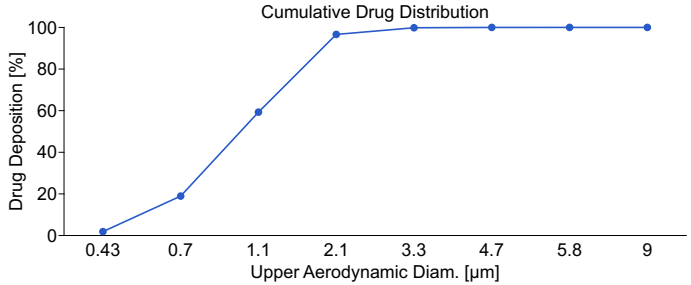




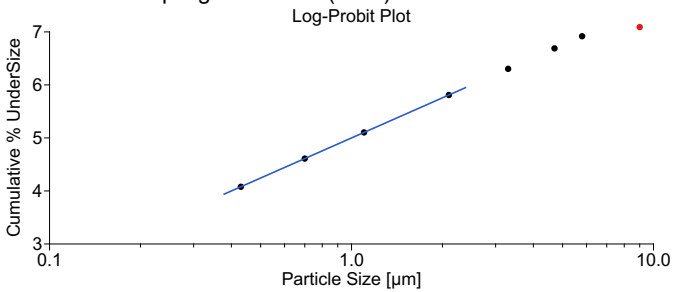
Run 5 Sampling Flow Rate (l/min) = 28.30

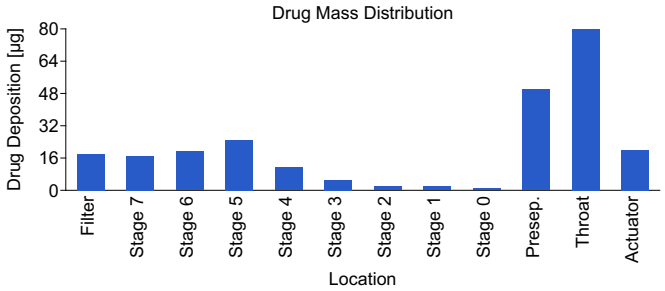


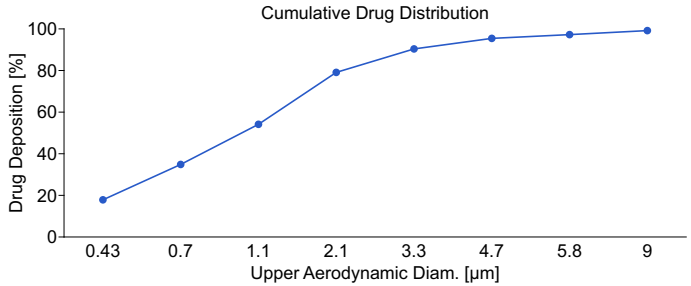




Run 6 Sampling Flow Rate (l/min) = 28.30







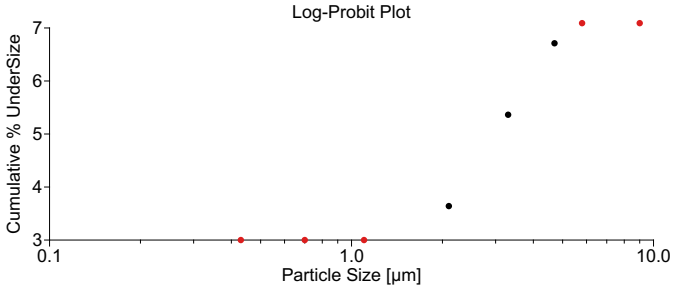
Graphical Summary for Cascade Impactor

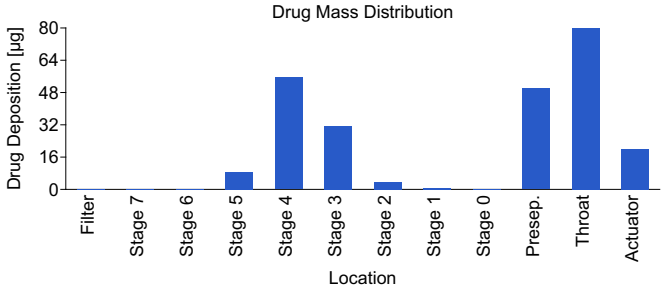
Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

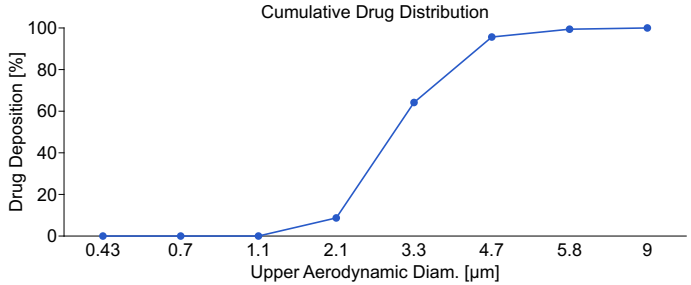
Other Info : Validation V3.1 - [File CI 1]

File Name: FILE CI 1.CPL

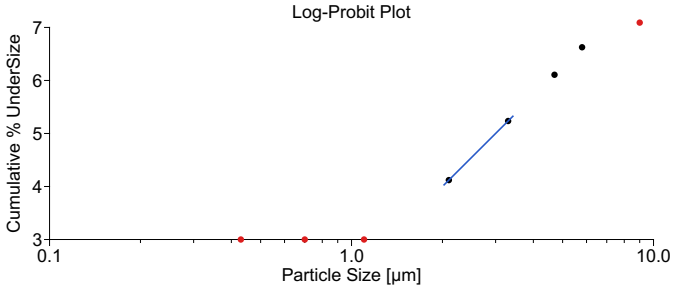
Run 7 Sampling Flow Rate (l/min) = 28.30

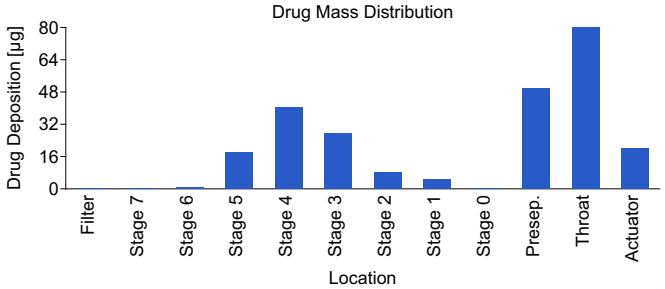


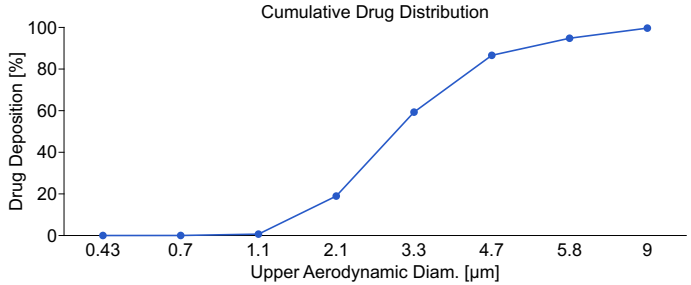




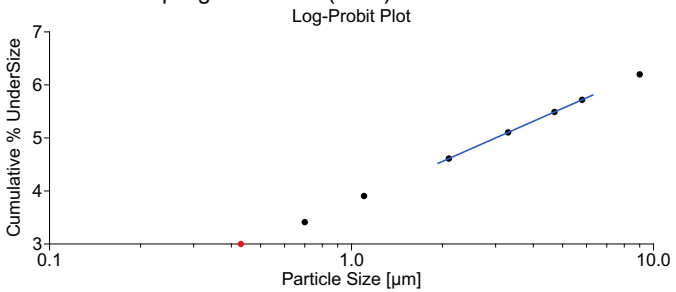
Run 8 Sampling Flow Rate (l/min) = 28.30

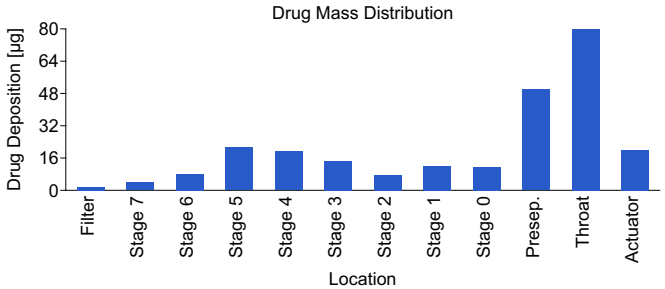


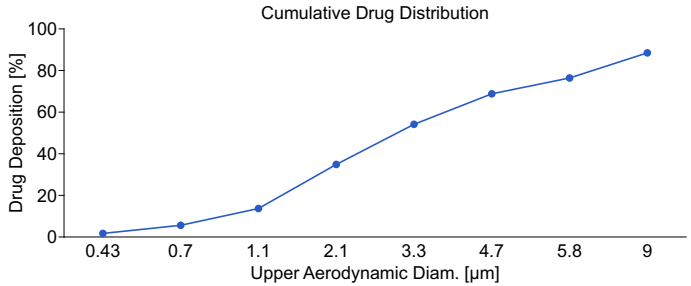




Run 9 Sampling Flow Rate (l/min) = 28.30







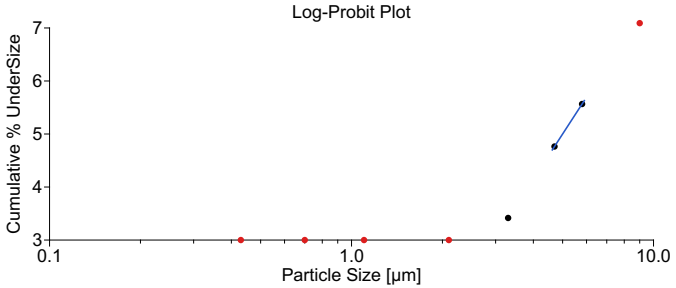
Graphical Summary for Cascade Impactor

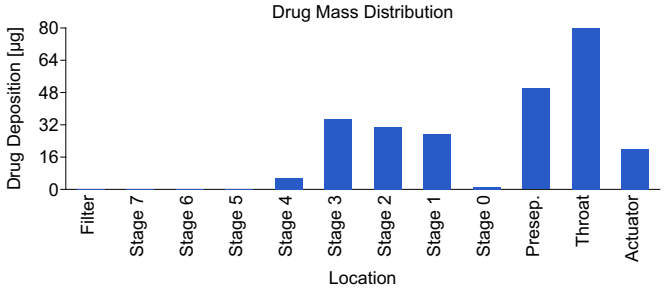
Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

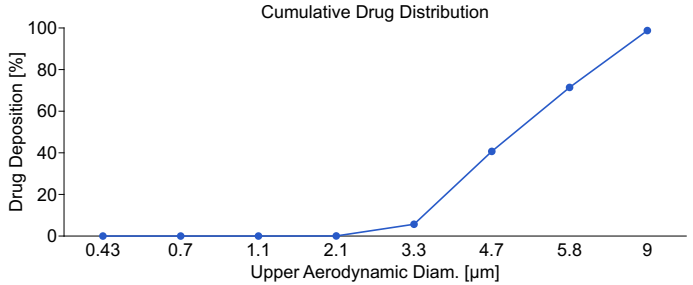
Other Info : Validation V3.1 - [File CI 1]

File Name: FILE CI 1.CPL

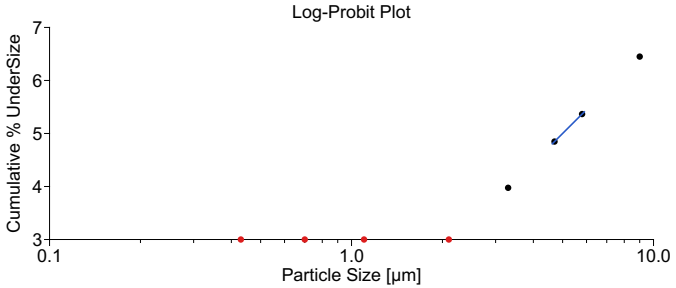
Run 10 Sampling Flow Rate (l/min) = 28.30

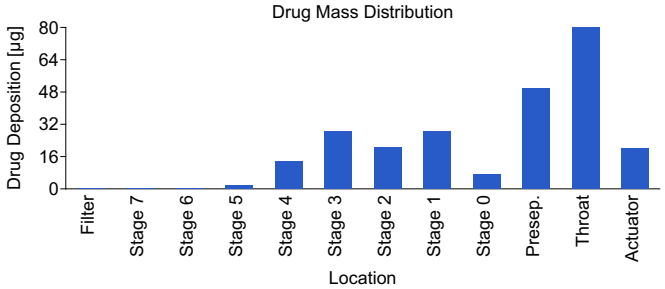


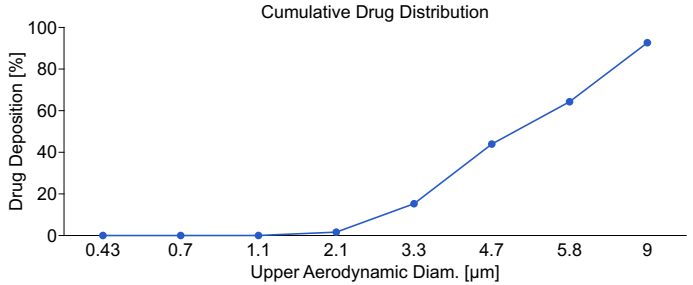




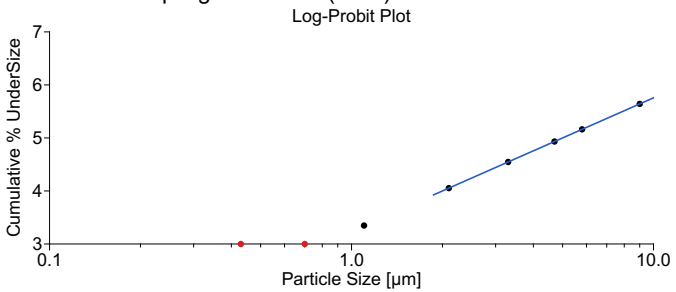
Run 11 Sampling Flow Rate (l/min) = 28.30

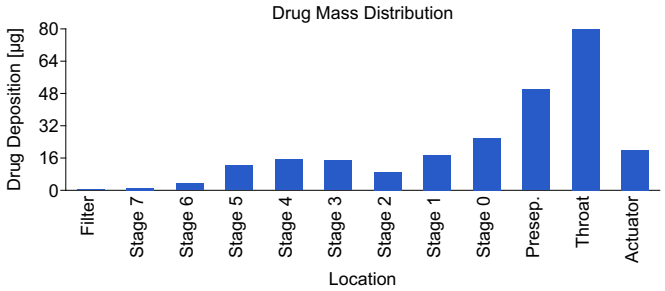


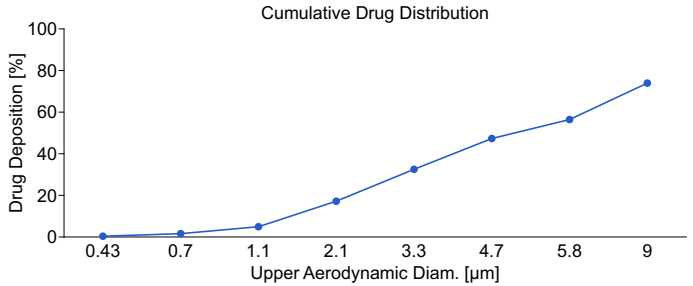




Run 12 Sampling Flow Rate (l/min) = 28.30







Tabular Summary for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

Other Info : Validation V3.1 - [File CI 1]

Fine Particle Dose Spec, Less than or Equal to 5.0um

File Name: FILE CI 1.CPL

Dose and Post Throat Particle Size Characterisation

Run No.	Dose Nos.	No. of Doses Sampled	Flow Rate Sampled	Metered [µg]	Delivered [µg]	Fine Particle [µg]	Fine Particle %	MMAD [µm]	GSD	R^2	n	Intercept	Slope
1		1	28.30	250.00000	230.00000	100.00000	43.47826	0.50000	1.30000	NA	1	NA	NA
2		1	28.30	249.99999	229.99999	99.99999	43.47826	0.50000	1.50000	1.00000	2	6.70951	5.67887
3		1	28.30	250.00001	230.00001	99.40136	43.21798	0.50000	2.50000	1.00000	3	5.75647	2.51294
4		1	28.30	250.00000	230.00000	100.00000	43.47826	1.00000	1.30000	NA	1	NA	NA
5		1	28.30	249.99999	229.99999	99.99927	43.47795	1.00000	1.50000	1.00000	2	5.00000	5.67887
6		1	28.30	250.00000	230.00000	96.04960	41.76069	1.00000	2.50000	1.00000	4	5.00000	2.51294
7		1	28.30	250.00001	230.00001	97.42332	42.35796	3.00000	1.30000	NA	1	NA	NA
8		1	28.30	250.00000	230.00000	89.61384	38.96254	3.00000	1.50000	1.00000	2	2.29049	5.67887
9		1	28.30	250.00002	230.00002	71.14048	30.93064	3.00000	2.50000	1.00000	4	3.80102	2.51294
10		1	28.30	249.99999	229.99999	49.99999	21.73913	5.00000	1.30000	1.00000	2	-1.13437	8.77629
11		1	28.30	250.00000	230.00000	50.00000	21.73913	5.00000	1.50000	1.00000	2	1.03064	5.67887
12		1	28.30	249.99999	229.99999	50.00000	21.73913	5.00000	2.50000	1.00000	5	3.24353	2.51294
Mean		-	28.30	250.00000	230.00000	83.63565	36.36333	2.37500	1.76667	1.00000	2.41667	3.52192	4.61595
SD		-	0.00	0.00001	0.00001	21.84625	9.49837	1.86017	0.54828	0.00000	1.31137	2.47696	2.22256
%RSD		-	0.00	0.00000	0.00000	26.12074	26.12074	78.32289	31.03447	0.00000	54.26368	70.32983	48.14961

Data Input by: Date: Data Checked by: Date:

Group Results for Cascade Impactor

Date of Analysis:	13-Mar-10	**Storage Conditions**	Number of doses/run =1
Analyst:	D Lewis	**Formulation Details:**	
Instrument Serial No.	CI Data Validation	**Drug/Dose**	Other Parameters
Device Details:		**Component 1**	Device: 20µg
Name	Flow Rate 28.3l/min	**Component 2**	Throat: 80µg
Device ID Code	MMAD = 0.5 1 3 and 5µm	**Component 3**	Stage 1: 50µg
Batch Number	GSD = 1.3 1.5 2.5	**Component 4**	Delivered Dose*: 230µg
Sample Age	Total stage mass = 100µg	**Component 5**	* uniformity of DD test

Other Info : Validation V3.1 - [File CI 1]

File Name: FILE CI 1.CPL

		Run 1	Run 2	Run 3	Run 4	Run 5	Run 6	Run 7	Run 8	Run 9	Run 10	Run 11	Run 12	Mean	SD	%RSD
Group 1																
From 0.00 µm	Particle Dose [µg]	99.58783	95.63218	77.53165	50.00000	50.00000	50.00000	LOD	LOD	11.52683	LOD	LOD	3.95040	54.77861	35.28923	64.42156
To 1.00 µm	Particle Fraction [%]	43.29906	41.57921	33.70941	21.73913	21.73913	21.73913	LOD	LOD	5.01166	LOD	LOD	1.71756	23.81679	15.34315	64.42156
Group 2																
From 1.00 µm	Particle Dose [µg]	0.39981	4.33639	15.95362	49.58782	45.63219	27.53165	LOD	LOD	21.37929	LOD	LOD	11.91512	22.09199	17.99730	81.46528
To 2.00 µm	Particle Fraction [%]	0.17383	1.88539	6.93636	21.55992	19.84008	11.97028	LOD	LOD	9.29534	LOD	LOD	5.18049	9.60521	7.82491	81.46528
Group 3																
From 2.00 µm	Particle Dose [µg]	0.01236	0.01981	3.98822	0.39149	4.03090	10.94152	43.18954	34.13448	17.09389	LOD	LOD	12.99401	12.67962	15.04010	118.61630
To 3.00 µm	Particle Fraction [%]	0.00537	0.00861	1.73401	0.17021	1.75257	4.75718	18.77806	14.84108	7.43212	LOD	LOD	5.64957	5.51288	6.53917	118.61630
Group 4																
From 3.00 µm	Particle Dose [µg]	0.00000	0.01161	1.36434	0.02069	0.29874	5.01210	36.35697	26.09964	12.32257	16.88965	18.71817	11.52024	10.71789	11.95372	111.53048
To 4.00 µm	Particle Fraction [%]	0.00000	0.00505	0.59319	0.00900	0.12989	2.17918	15.80738	11.34767	5.35764	7.34333	8.13833	5.00880	4.65995	5.19727	111.53048
Group 5																
From 4.00 µm	Particle Dose [µg]	0.00000	0.00000	0.56352	0.00000	0.03743	2.56432	11.06635	13.51421	8.81790	30.24793	20.89568	9.62023	8.11063	9.74435	120.14292
To 5.00 µm	Particle Fraction [%]	0.00000	0.00000	0.24501	0.00000	0.01628	1.11492	4.81146	5.87574	3.83387	13.15128	9.08508	4.18271	3.52636	4.23667	120.14292

Data Input by: Date: Data Checked by: Date: